

YOLO Detection Dataset Spectral Comparison

Particles are exact YOLO label intervals.

Dataset noise is made of FFT windows that do not overlap any label.

Standalone Noise is a single external reference, shown once in neutral color.

Datasets:

- YOLO v3 old pipeline: P1/yolo_dataset_v3
- P0 C1 Hamming: P0/data/dataset_hamming_c1_yolo
- P0 C1 clean + 7-80 kHz: P0/data/dataset_hamming_c1_clean_filt_yolo_trainval
- Standalone Noise reference: P0/data/Noise

Coverage summary:

P0 C1 clean + 7-80 kHz test: signals=578, labels=895, mean coverage=15.7%, p95=36.6%

P0 C1 Hamming test: signals=578, labels=6905, mean coverage=66.6%, p95=93.9%

YOLO v3 old pipeline test: signals=354, labels=384, mean coverage=7.0%, p95=14.1%

10-40 kHz and overlap summary:

YOLO v3 old pipeline 2um: particle=71.0%, dataset-noise=19.1%, Noise reference=19.0%, dominant overlap=False

YOLO v3 old pipeline 4um: particle=93.0%, dataset-noise=19.2%, Noise reference=19.0%, dominant overlap=False

YOLO v3 old pipeline 10um: particle=72.2%, dataset-noise=19.2%, Noise reference=19.0%, dominant overlap=False

P0 C1 Hamming 2um: particle=30.2%, dataset-noise=8.7%, Noise reference=19.0%, dominant overlap=False

P0 C1 Hamming 4um: particle=51.5%, dataset-noise=8.9%, Noise reference=19.0%, dominant overlap=False

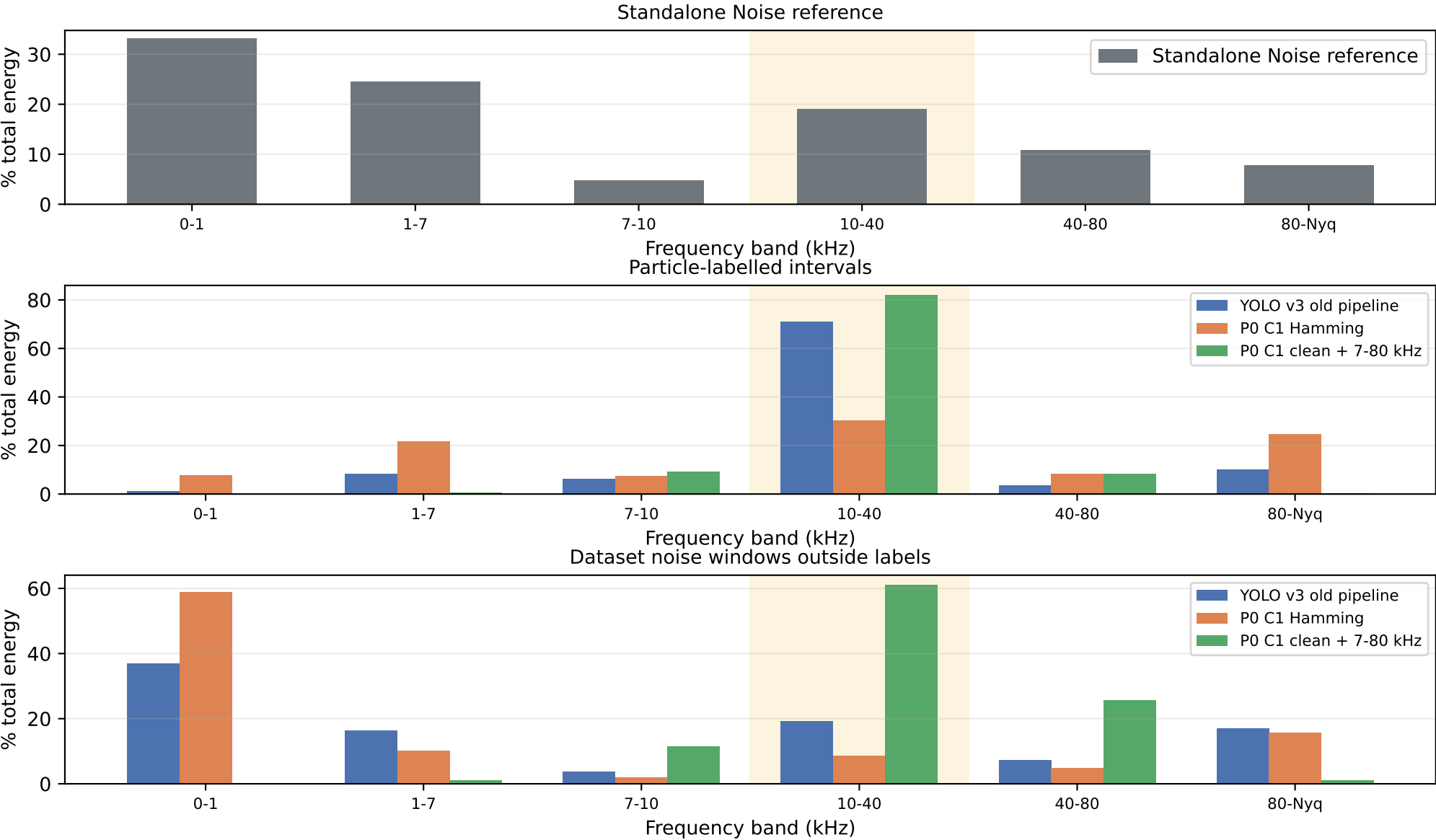
P0 C1 Hamming 10um: particle=21.5%, dataset-noise=8.9%, Noise reference=19.0%, dominant overlap=False

P0 C1 clean + 7-80 kHz 2um: particle=81.9%, dataset-noise=61.0%, Noise reference=19.0%, dominant overlap=True

P0 C1 clean + 7-80 kHz 4um: particle=96.5%, dataset-noise=61.0%, Noise reference=19.0%, dominant overlap=True

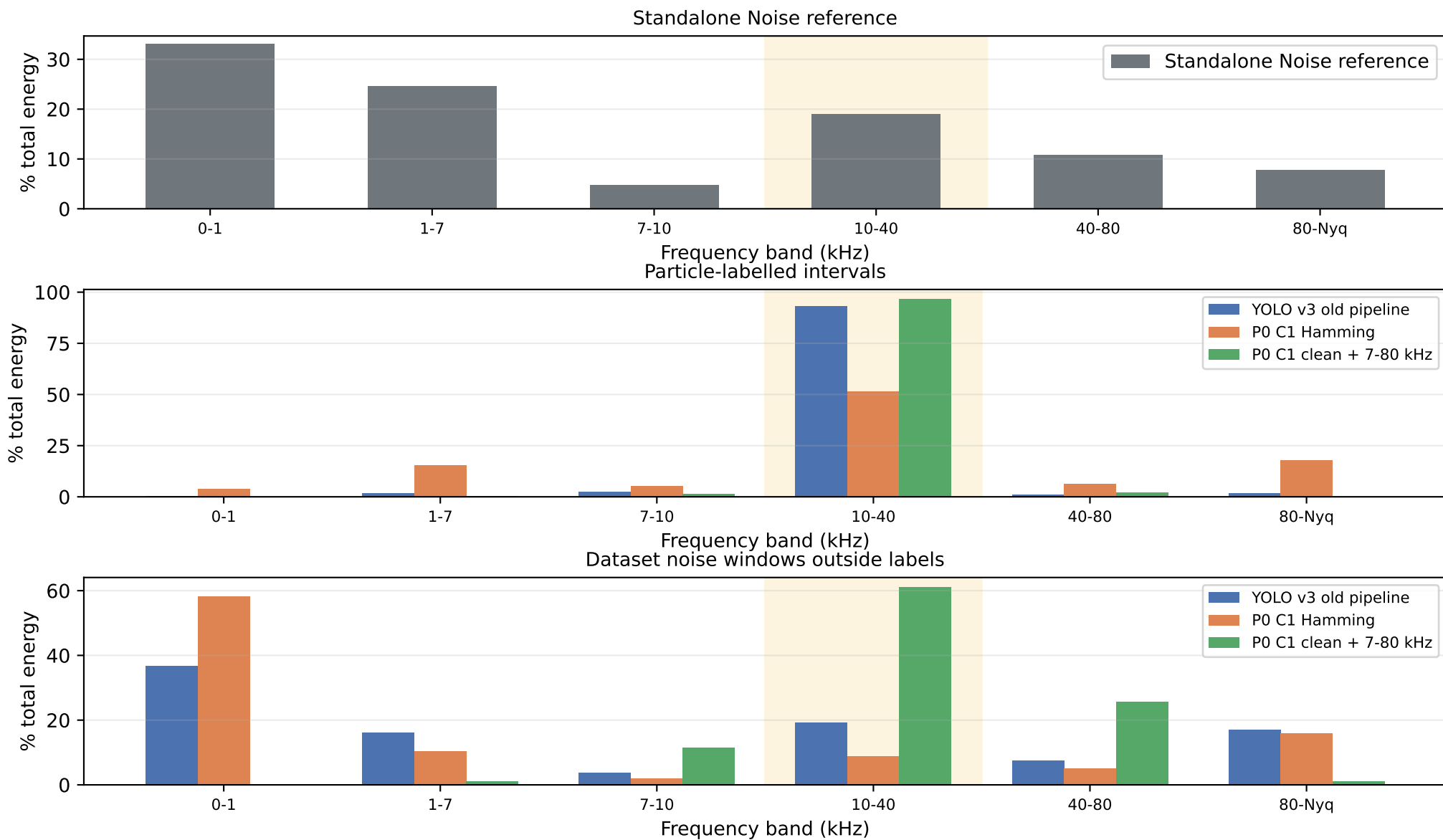
P0 C1 clean + 7-80 kHz 10um: particle=78.8%, dataset-noise=61.0%, Noise reference=19.0%, dominant overlap=True

2um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



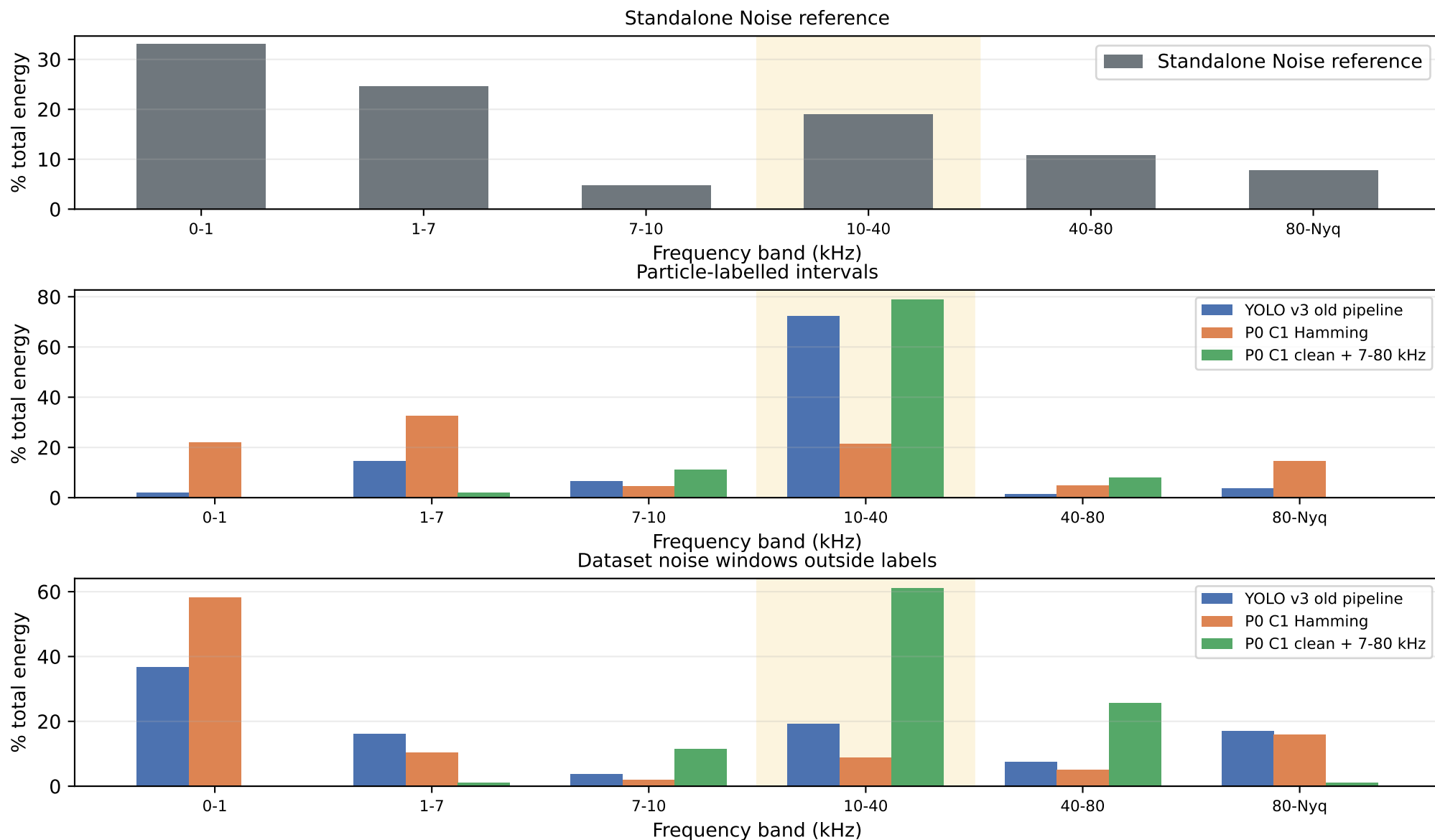
Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	71.0%	19.1%	19.0%	7.0%
P0 C1 Hamming	30.2%	8.7%	19.0%	66.6%
P0 C1 clean + 7-80 kHz	81.9%	61.0%	19.0%	15.7%

4um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	93.0%	19.2%	19.0%	7.0%
P0 C1 Hamming	51.5%	8.9%	19.0%	66.6%
P0 C1 clean + 7-80 kHz	96.5%	61.0%	19.0%	15.7%

10um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	72.2%	19.2%	19.0%	7.0%
P0 C1 Hamming	21.5%	8.9%	19.0%	66.6%
P0 C1 clean + 7-80 kHz	78.8%	61.0%	19.0%	15.7%